Master RC Actuator Controls and Slave Information

Testing guide revision 1 — October 3, 2026

> This guide describes unreleased testing firmware. These features are not part

> of the current public Installer release. Use only the matching testing builds

> supplied for your Master and slave controllers.

RC Actuator Controls temporarily give one RC stick control of a servo, a saved servo group, or a two-axis mechanism. The selected RC button toggles that control on and off, so the operator does not have to hold the button while moving the stick.

The Master can also read saved device information from compatible DL_Slave and DL_AP controllers. After an update, the Master shows actual output names, servo numbers, output types, saved groups, and assigned sequence shortcuts.

Safety first

- Disconnect mechanical linkages while establishing servo endpoints.
- Keep drive wheels raised and disconnect motor power during initial testing.
- Center the selected stick before toggling actuator control on.
- Confirm Device ID, Servo #, Minimum, Maximum, and Home before Live Test.
- Test one mechanism at a time until direction and travel are verified.
- Keep a physical power switch within reach.

The Master locks drive outputs during RC Actuator Live Test. Normal runtime operation retains the usual RC signal-loss, drive-stop, dome-stop, and centered-stick protections.

Compatible testing firmware

- DroidLink Master V2.2.0 testing firmware
- DroidLink Slave V2.4.0 testing firmware
- DroidLink_AP V2.4.0 testing firmware

Older devices remain usable for normal commands. They cannot provide saved names and lists to the Master, so they appear as not updated or unsupported.

Update Slave Info

Use this after changing output names, output types, servo groups, or saved sequence assignments on DL_Slave or DL_AP.

1. Finish and save the changes on the slave controller.
2. Return the droid to normal operation.
3. Enter the Master's normal runtime Web Config mode.
4. Open the Master main screen.
5. Select **Update Slave Info**.
6. Wait for the status message to report completion.
7. Open **Device Status** to review firmware and saved item counts.

The update is manual by design. The Master does not download every complete device configuration on every boot. It asks one detected device at a time and saves the validated results locally.

Drive and dome input processing continues during the update. Starting an update ends any active RC Actuator session first. A compatible slave pauses only its RC live-actuator input while its information is being read; its normal configuration is not copied to or changed by the Master.

If a device is off, out of range, or running older firmware, its last saved information may remain visible as a saved cache. Turn the device on and select **Update Slave Info** again after correcting the problem.

What the Master learns

For each compatible device, the Master saves:

- Device ID, name, role, firmware version, and output controller type
- Servo/output numbers, names, output types, and enabled state
- Saved servo group names and group membership
- Saved sequence names and their assigned DroidLink shortcuts
- Supported actuator and device-information capabilities

The slave remains the authority for calibration, endpoints, direction, output mode, groups, and sequences. The Master uses saved information to make its selection screens easier and safer; it does not replace slave configuration.

Servo numbering

Servo # is zero-based. The first configurable output is Servo #0, the second is Servo #1, and so on. This is true for PCA9685 and Pololu Maestro controllers.

After **Update Slave Info**, Servo # fields offer the saved output name and type. Manual number entry remains available for older firmware and advanced testing.

The Servo # is DroidLink's logical output number. It is not necessarily the printed PCA9685 channel or Maestro controller's local channel number.

Create an RC Actuator control

1. Open **RC Controls** on the Master.
2. Open **Global** or the desired RC profile.
3. Open **Actuators**.
4. Select **Add Actuator Control**.
5. Name the control and select its toggle button and stick axis.
6. Select the control mode, Device ID, and target.
7. Set motion limits, Home, speed, and end behavior.
8. Add an optional companion command if needed.
9. Save RC Inputs from the RC Controls main screen.

Global controls are available in every profile. A control saved in Profile 1 through Profile 4 is available only while that profile is active.

Toggle operation

1. Center the selected stick.
2. Press the configured toggle button once.
3. Move the stick to control the mechanism.
4. Center or stop the mechanism as appropriate.
5. Press the same button again to end control.

When control ends, the selected stick returns to its normal drive or dome function only after it is centered. This prevents unexpected movement when ownership changes.

Target types

Single Servo

Controls one logical Servo # on the selected Device ID.

Saved Servo Group

Controls every enabled servo in a group saved on the selected slave. After a device-information update, the field offers groups that exist on that device.

Two-Axis Servo Pair

Controls two independent servos from the horizontal and vertical directions of one stick. This is useful for two-servo holoprojectors and pan/tilt devices.

Control modes

Jog (Hold to Move)

Moving away from center moves the servo. Returning to center holds the current position. Stick distance controls movement rate. Use Jog for lifters and arms.

Follow Stick Position

Servo position follows stick position. Full travel selects Minimum or Maximum; center selects Centered/Home.

Two Positions (Min / Max)

One stick direction selects Minimum and the other selects Maximum. This suits doors, latches, and other two-position mechanisms.

Continuous Servo (360°)

Stick direction selects forward or reverse, stick distance controls speed, and center commands stop. The slave output must be configured as a 360° servo.

360° + Position

One axis controls a positional servo while the other controls a 360° servo. This supports rotation plus a separate position or steering function.

Minimum, Maximum, Home, and Speed

- **Minimum** and **Maximum** limit the portion of the slave's calibrated travel available to RC control.
- **Home** is the return or centered position used by the selected mode and by **Return Home**. It does not change saved endpoints.
- **Speed** limits positional movement. A continuous servo uses stick distance for requested rotation speed within its saved range.
- **Reverse stick direction** reverses increasing position or rotation.

For a mechanism closed at the saved closed endpoint, Home is normally 0%.

When control ends

- **Hold Position** leaves a positional servo where it stopped.
- **Return Home** moves a positional servo to Home.
- **Disable Output** stops sending the positional servo signal.
- **Stop (360°)** sends neutral/stop to a continuous servo.

Available choices depend on the mode. A 360° servo must stop when control ends.

Companion command

The optional Companion Command runs when the toggle changes actuator control. It may be any valid DroidLink command, not only audio.

Examples include toggling a door before moving an arm, starting lighting, opening a panel while taking control of a lifter, or running a saved sequence. Enter the complete command. The device catalog offers assigned saved-sequence shortcuts as suggestions, while manual command entry remains available.

Live Test

Live Test lets the physical RC controller operate configured actuators without leaving the RC Controls page.

- Finish the current field edits before arming.
- Verify the target and limits before selecting **Arm Test**.
- Drive outputs remain locked throughout the test.
- Stop the test before editing device configuration or leaving the page.

Live Test does not enable the drive motors.

Master Sequences

After **Update Slave Info**, assigned slave sequence shortcuts appear as suggestions for a Master Sequence command step. The saved Master Sequence uses the normal DroidLink command at runtime.

MarcDuino limitation

DL_AP in MarcDuino mode can report supported saved sequence shortcuts, but it does not expose configurable servo endpoints, saved servo groups, or RC live actuator control. DroidLink passes MarcDuino commands through; it does not own the endpoints and output state needed for safe live positioning.

Troubleshooting

No compatible devices were found

- Confirm the slave is powered and paired to the correct Master MAC.
- Confirm its Device ID and MAC are in Master System Setup.
- Confirm both devices use matching catalog-capable testing firmware.
- Refresh Device Discovery and try Update Slave Info again.

A device shows saved cache

The Master has last-known information, but that device did not complete the most recent update. Check power, range, pairing, and firmware, then update.

A servo or group is missing

Save it on the slave, return to Master runtime Web Config, and update again. Disabled or omitted outputs are not offered as normal actuator selections.

An actuator will not take control

- Center the selected stick before pressing the toggle.
- Confirm the correct RC profile is active and the device is online.
- Confirm the target output is enabled and calibrated.
- Check that the same RC input is not assigned to a conflicting action.

Stick direction feels backward

Use **Reverse stick direction**. Horizontal and vertical reversal are separate for a two-axis control.